Mehdi Eskandari-Ghadi

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EDUCATION

Civil Engineering Ph.D. (GPA 4.0), Engineering Science Focus **2016 – 2021**
University of Colorado Boulder, Boulder, CO

TECHNICAL SKILLS

Programming	Python, TypeScript/JavaScript, Git, VBA, MATLAB, AutoLISP
Platforms & Data	Palantir Foundry (Ontology, Pipeline Builder, Contour, Workshop, OSDK, AIP), Spark, Polars, Pandas
Engineering	AutoCAD, PLSCADD, L-PILE, ABAQUS/LS-DYNA
Standards & Tools	NESC, GO-95, FERC-881, Microsoft Office Suite, LaTeX

ENGINEERING EXPERIENCE

Reliability & System Modeling Engineer / System Modeling Principal Engineer — PacifiCorp **Jan 2025 – Present**

- Developed a reusable Python library implementing physics-based, IEEE-compliant thermal rating models, numerically solving the nonlinear heat balance through Runge-Kutta integration and Newton-Raphson root-finding, vectorized at scale for static, dynamic, and transient line and equipment ratings. Implemented seasonal ambient temperature analysis over 30 years of daily weather data across thousands of miles of power lines.
- Designed, built, and deployed the company's Foundry-based facility rating data management system end to end with full-stack TypeScript and Python, owning its architecture, implementation plans, and delivery, and migrating prototype FERC-881 line-rating and FAC-008 equipment-rating business logic into production. The system spans backend and queue-based processing pipelines, user-facing applications, audit preparation, and interdepartmental data collection, delivered concurrently with testing and production adoption.
- Built an enterprise, operations-scale Foundry ecosystem that analyzes the transmission system from the equipment up: it links equipment specifications to facilities, runs physics-based computations of weather-adjusted equipment ratings and geometric mutual-impedance analyses, and identifies each facility's range of operation, which operations engineers run facilities within and protection and control engineers set their controls around. It runs on an Ontology redesigned for semantic clarity and workflow object types, with 100+ Actions and Functions and 30+ Pipeline Builder pipelines, and interfaces with external systems through automations, maps, graphs, and other UI elements.
- Designed backend Python data pipelines in Spark and Polars to ingest, parse, and remediate enterprise-scale, imperfect legacy data from multiple database and spreadsheet sources, automating cleanup and multi-source merging to reconcile records to production quality standards. Reviewed and debugged other engineers' data pipeline logic and data processing code.
- Built reliability safeguards: unit and end-to-end tests, dataset primary key checks, validation rules, monitoring panels flagging missing object links, and pre-run validation previews for resource-consuming tasks.
- Architected a reusable, git-inspired version-control framework for system analysis lifecycle management in Foundry, writing its design, information flow, and edge-case handling before implementation: facility analyses are cloned, modified, and published in stages through department-specific review checkpoints, with automated resource tracking and validation.
- Directed AI coding agents with repository and Foundry Ontology access by sequencing incremental implementations and large migrations to write and edit functions, entire features, and React-based custom Workshop widgets through a full commit, tag, and pull-request workflow, coordinating agent-driven changes across multiple interdependent repositories while maintaining version and compatibility consistency.
- Owned end-to-end UX design in Foundry Workshop, continuously identifying friction points and iterating on layout and information architecture to keep complex, multi-source engineering workflows simple and intuitive for end users, with shared layouts, color schemes, and reusable components (map, facility table, metric cards) across all modules. Drove adoption by training PacifiCorp engineers, writing user guides and code documentation, and holding weekly feature-update meetings.
- Built a reusable Python report-processing library and used it to power LLM-based reporting via Palantir AIP, generating narrative reports on system modification changes and their system implications, in active use to focus administrative review across data spread over legacy tables, files, engineering drawings, and internal apps.
- Designed and built a streamlined FAC-008 substation equipment rating tool from scratch in VBA/Excel, applying IEEE equipment-rating knowledge to compute normal and emergency ratings and rank limiting elements from company standards and nameplate specifications.
- Deployed AutoCAD experience and problem breakdown skills to implement AutoLISP one-line-drawing parsers. The parsers identify transmission facilities and equipment, deduce line segment and equipment configurations, and export structured data to reconcile database records with engineering drawings.

Transmission Engineer I/II — PacifiCorp **Jan 2023 – Jan 2025**

- Utilized VBA and Excel programming alongside transmission line rating engineering knowledge to design and build an auditable, automated FERC-881 steady-state line-rating analysis tool from scratch. This tool integrates PLS-CADD structure data, conductor properties, and historical weather data with CIGRE-referenced seasonal rating methodology to compute normal and emergency ratings and critical-span summaries. It streamlined grid line data extraction, reducing engineering hours threefold, and initialized the Foundry datasets for future system migration, validation, and automation.
- Designed new transmission lines and rebuild and maintenance projects for existing lines in six states with PLSCADD, L-PILE, NESC, and GO-95, collaborating with teams of engineers, managers, field crews, and legal.
- Scoped new transmission lines with other engineering disciplines, and applied structural and geotechnical engineering to custom-engineer transmission structures and drilled pier foundations for special needs per ASCE, ACI, and FHWA.
- Overhauled the company's transmission structure library from legacy templates into modular, component-based PLS-POLE libraries, and designed a new class of circuit-swap structures for repositioning circuits on a double-circuit extra-high-voltage line with minimal increase to structure size, vetted across engineering and field operations to meet electrical and structural requirements.
- Developed automated Excel tools for deep foundation design, construction package generation, transmission pole design aids, and company sag-tension standards for tree-wire.
- Designed and reviewed drawings for tubular steel poles and drilled piers to withstand vertical and lateral loads, and improved company standards and procedures through detail-oriented QAQC reviews.
- Performed detail-oriented technical review of transmission standards, contributing to their development, and initiated process improvement pipelines for transmission design processes and documentation.

RESEARCH EXPERIENCE

Graduate Research Assistant / Ph.D. Candidate / Postdoctoral Researcher **Jun 2018 – Jan 2023**

University of Colorado Boulder, Boulder, CO

- Collaborated with a team of senior researchers at the Lawrence Berkeley National Laboratory to investigate and develop mechanistic models for subcritical crack growth healing, applying fundamentals of fracture mechanics, fluid transport, and boundary element methods to develop computational software that couples fields of mechanics and physics, and gave regular presentations on crack propagation to LBNL BES scientists with backgrounds in mechanics, physics, and chemistry.
- Investigated multiphysical, multiscale solid-fluid interactions (adsorption, surface tension, solvation pressure) in porous solids and developed thermodynamics-poromechanics-based MATLAB software modeling adsorption-induced deformation of microporous material, running parametric studies to test model sensitivity.
- Authored 4 peer-reviewed publications in JMPS, IJSS, and EFM.

EMI Virtual Conference 2021 and EMI Conference 2022 **May 2021 and May 2022**

- Applied illustrative and verbal communication skills to deliver presentations titled "A multi-scale theory explaining the initial shrinkage of micro-porous solid upon gas adsorption" and "A Multiphysical Surface-Force Based Fracture Theory for Subcritical Crack Growth in Surface-Reactive Environments" to engineering and academic audiences.
- Won runner-up in the 2021 Student Poster Competition for a poster on sorption-induced deformation of porous material.

TEACHING EXPERIENCE

Teaching Assistant / Tutor **2014 – 2019**

University of Colorado Boulder, Boulder, CO & University of Tehran, Tehran, Iran

- Presented class lectures and laboratory teaching sessions, held office hours and exam preparation sessions, prepared and graded exams, homework, and projects, for courses of Advanced mechanics of materials, Dynamics, Structural Analysis II, and Hydraulics of open channels.
- Tutored students for the FE exam with a 100% success rate.

LEADERSHIP ROLES

Research Mentor **Jan 2022 – Jan 2023**

University of Colorado Boulder, Boulder, CO

- Trained new Ph.D. students on fundamentals of mechanics, written and oral skills, and technical research, and guided a research trainee through the modeling and simulation of beam bending in humidity actuators.

President/Co-president — Persian Student Organization, Boulder, CO **Aug 2018 – May 2019**

University of Colorado Boulder, Boulder, CO

- Proposed, secured funding, planned, and oversaw student events with industry, academic, business, and nonprofit partners.